EchoPulse

Smart Flood Monitoring and High-Efficiency Water Management System

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Green Radar Systems for Environmental Monitoring

.Introduction

The EchoPulse system aims to provide an innovative solution based on the use of radars and sensors to monitor water levels with high accuracy, while reducing energy consumption through intelligent operation. The system also contributes to analyzing water quality and its reuse potential, thus promoting the concept of sustainability and effective water resource management

. Definition Problem

Floods are among the most dangerous natural disasters that can cause significant damage to infrastructure and the environment.
Reports indicate that seasonal floods in Yemen in 2024 affected more than 420,000 people and destroyed infrastructure in 19 governorates. In the UAE, heavy rains caused financial losses of up to $3.4 billion. These figures demonstrate that floods in the Middle East pose a real threat to both people and the environment. Many traditional flood monitoring systems rely on the continuous operation of radar equipment, resulting in significant energy consumption. Furthermore, floodwater is often wasted,

despite its potential use in areas such as irrigation and agriculture if its quality is suitable.

Proposed Solution

This project proposes an intelligent system based on radars and sensors to monitor water levels in rivers or dams and detect floods early.

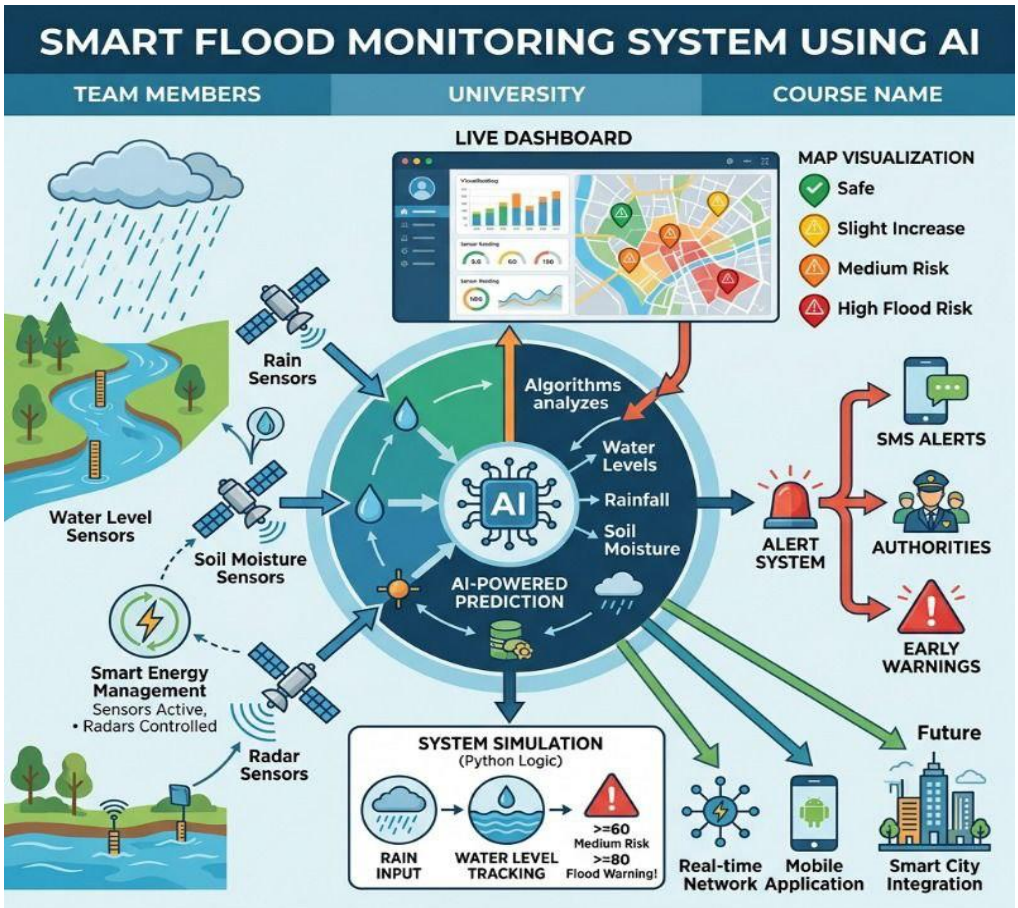


Figure (1): Diagram of an intelligent flood monitoring and forecasting system using sensors and artificial intelligence

The system divides the area into several monitoring zones, each containing a water level sensor and a radar for precise surface monitoring.

The low-power sensors operate continuously to measure the water level, while the radar is activated only when abnormal changes occur. This significantly reduces energy consumption.

Upon detecting a dangerous rise in the water level, the system sends an early warning, and excess water can be diverted to designated drainage channels. Water quality is then measured using specialized sensors to determine its suitability for irrigation and agricultural use.

The system's efficiency can be enhanced by using artificial intelligence and machine learning techniques to analyze data collected from water level sensors and radars, as well as environmental data such as rainfall, temperature, and weather conditions.

The model analyzes patterns and changes in water level over time, enabling the system to predict rising water levels and the likelihood of flooding hours or even days in advance. This

early prediction helps authorities take preventative measures, such as activating drainage systems or issuing early warnings to residents. In at-risk areas, this reduces potential damage and improves water resource management.

"As part of the EchoPulse flood monitoring system, the project includes an interactive map feature for at-risk areas. The map shows the water level in each area and is colored according to the risk level:

- Red: Very high water level → High risk
- Orange: Medium water level → Medium risk
- Yellow: Low water level → Low risk

The map shows the most affected countries and regions, and users can click on any area to view details such as the number of people affected, the number of deaths, and infrastructure losses.

This map helps authorities make quick decisions, such as issuing warnings or activating drainage systems, thus promoting smart and sustainable water resource management."

System Overview

The EchoPulse system consists of a set of sensors and radars connected to a central processing unit. The sensors continuously monitor the water level, while the radar is activated only when abnormal changes occur. The data is sent to a central system for analysis and displayed on an interactive dashboard, enabling users to make quick and effective decisions.

Functional Requirements

The system offers a range of essential functions, including:

- Continuous measurement of water levels using sensors
- Radar activation upon detection of abnormal changes
- Data analysis to determine flood risk
- Immediate alerts in case of danger
- Data display on an interactive dashboard
- Water quality analysis using dedicated sensors.

Non-Functional Requirements

The system must meet the following characteristics:

- High accuracy in water level measurement
- Low energy consumption
- Ability to operate continuously (24/7)
- High response speed in case of danger
- Secure data protection system
- Scalability to monitor multiple areas

Radar Type and Operating Scenario

The system relies on the use of an FMCW (Frequency Modulated Continuous Wave) radar. This type is characterized by its high accuracy in measuring distances and its low power consumption compared to other types of radars.

The radar transmits electromagnetic waves towards the water's surface and then receives the reflected signal. By analyzing the difference between the transmitted and received signals, the distance between the radar and the water's surface can be calculated, thus determining the water level.
The system's operating scenario is as follows:

1. Sensors continuously monitor the water level.
2. If an abnormal rise in the water level is detected, the radar in that area is activated.
3. The radar accurately measures the distance to the water surface.
4. The data is analyzed to determine the presence of a flood risk.
5. If a risk is present, an alarm is issued and the water drainage system is activated.

System Architecture

The system consists of several key components working together to achieve intelligent water monitoring:

- Water level sensors
- Water surface monitoring radars
- Central processing unit for data analysis
- Data transmission and storage system
- Dashboard for displaying information
- Early warning system
- Water quality sensors

The sensors send data to the processing unit where it is analyzed and the appropriate decision is made. The results are displayed on an interactive control panel that helps users easily understand the system status.

The system also relies on the concept of a smart radar network, where each radar unit acts as a node within an Internet of Things (IoT)

network. These units communicate with each other and with the processing center via low-power communication technologies, enabling continuous and efficient data transfer.

This approach helps cover large areas and monitor multiple sites simultaneously while maintaining low energy consumption.

Water Quality and Pollution Monitoring

In addition to monitoring water levels, the system includes sensors to measure water quality, such as:

- pH sensor
- Turbidity sensor
- Temperature sensor

These sensors help determine the suitability of water for irrigation and agriculture. The system can also detect water pollution if abnormal readings appear.

This contributes to environmental protection and more efficient water resource management.More efficiently.

System Workflow

The system operates according to the following steps:

- Sensors continuously monitor the water level.
- When an abnormal rise is detected, the radar is activated.
- The radar measures the water level with high accuracy.
- The data is sent to the processing unit for analysis.
- In case of danger, an early warning is issued.
- Water drainage systems can be activated automatically.

Database Design

The system includes a database for storing information, consisting of the following tables:

- ✓ WaterLevels (for storing water level readings)
- ✓ Alerts (for storing alerts and hazards)
- ✓ Locations (for determining the locations of sensors and radars)
- ✓ Users (for storing user data)

User Guide

The user can easily interact with the EchoPulse system through the dashboard as follows:

1. Log in using username and password
2. Access the interactive map that displays the status of the affected areas In real time
3. Identify the risk level using colors:
   - Red = High risk
   - Orange = Medium risk
   - Yellow = Low risk
4. Click on any area to view details (number of affected, fatalities, losses)
5. Receive instant alerts on your mobile phone + take quick actions (activate drainage, evacuate, etc.)

Testing

- The system was tested to ensure its efficiency through:
- Testing the accuracy of the sensors in measuring water levels
- Testing the accuracy of the radar in determining distance
- Testing the system's response speed in the event of a hazard
- Testing the correct transmission of alerts

- Testing the system's operation under different conditions

Security

- The system ensures data protection through:
- Using system login mechanisms for users
- Defining access permissions for each user
- Protecting data during transmission and storage
- Preventing unauthorized access to the system

Maintenance

- The system requires periodic maintenance to ensure its continued operation, which includes:
- Regular inspection of sensors and radars
- Software updates and performance optimization
- Repairing malfunctions when they occur
- Replacing damaged equipment

Sustainability benefits

The system offers several sustainability benefits, including:

❖Energy savings:

Radars are only activated when needed, reducing energy consumption compared to systems that operate continuously.

❖Flood damage reduction:
The system helps detect floods early and send warnings before the danger occurs.

❖Sustainable water management:
Excess water from floods can be utilized for agriculture after its quality is analyzed.

After directing the excess water to dedicated channels, the system analyzes the water quality using specialized sensors such as a pH sensor and a turbidity sensor.

If the water quality is suitable, it can be used for agricultural irrigation or stored as an additional water source.

However, if signs of pollution are detected, the system sends an alert to the relevant authorities to take the necessary measures.

❖Pollution Monitoring:
The system helps detect changes in water quality and thus contributes to environmental protection.

The system is based on the principle of intelligent radar operation (Event Activation Driven), where low-power sensors continuously monitor water levels, while the radar is activated only when an abnormal change in readings is detected.

This method significantly reduces energy consumption compared to traditional systems that operate continuously.

Energy Saving Strategy

EchoPulse relies on intelligent operation (Driven-Event)
Saving up to 75% of energy, as the radar remains in standby mode until it is automatically activated only when needed.

Expected Results

The system is expected to achieve the following results:

1. Accurate water level monitoring
2. Early flood detection
3. Reduced energy consumption through intelligent radar operation
4. Improved management of excess water

5. Monitoring water quality and pollution

The EchoPulse system can be applied in many different environments, such as:

- River monitoring
- Dam and reservoir management
- Flood-prone areas
- Cities near valleys

This contributes to enhancing water resource management and reducing environmental risks.

Innovation Highlights

What makes EchoPulse different and innovative?

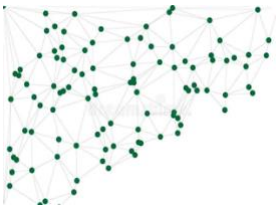
- ✓ The first smart green radar system operating on the principle of Driven-Event Activation, low-power sensors operating 24/7 + FMCW radar that only activates when an abnormal altitude is detected Energy savings of up to 75% compared to conventional systems.

- ✓ Integrating radar with artificial intelligence for early prediction: The model predicts flooding 6-24 hours in advance using data on: Level + Rainfall + Temperature + Humidity.

- ✓ Smart reuse of excess water (new hackathon feature): After directing water to dedicated channels → Real-time analysis of pH + Temperature + Turbidity → If suitable, it is automatically diverted for agricultural irrigation or storage.

- ✓ Interactive color map + Dashboard displaying real-time risk levels (red/orange/yellow) with the ability to tap to view the number of casualties, deaths, and losses.

- ✓ Smart IoT radar network: Each radar unit operates as an independent node and communicates with each other using Power-Low technology, providing wide coverage at very low cost and energy consumption.



- ✓ Green Radar + Sustainability in One System Combining green radar, circular water management, and environmental protection in one integrated solution

Conclusion

The EchoPulse project now offers a smart and sustainable solution for flood monitoring and water resource management. By integrating radar with sensors and data analysis technologies, the system can provide accurate water level monitoring while reducing energy consumption. The system also contributes to utilizing excess water and analyzing its quality for agricultural use, thus promoting the concept of sustainable water management.